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ORIGINAL PAPER

Tree-pulling experiment: an analysis into the mechanical stability of rain trees

Lai Fern Ow · F. R. Harnas · I. G. B. Indrawan · A. Sahadewa · E. K. Sim · H. Rahardjo · E. C. Leong · Y. K. Fong · P. Y. Tan

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Abstract Tree-pulling experiments were conducted in Singapore involving 20 rain trees (*Samanea saman*) growing in four different soil types (1) structural soil with 80% granite chips and 20% sandy loam soil, (2) structural soil with 50% granite chips and 50% sandy loam soil, (3) in situ soil and (4) top soil. The trees were pulled over with a winch attached to the stem at a standard height of 1.3 m and the force required to uproot or break the trees were recorded. The physical above and below ground characteristics of the trees were also measured. All 20 trees in this study failed via uprooting without any stem fracture. Analysis of the data showed that the maximum resistive bending moment (BM_{max}) was positively correlated with the overall size of the root plate, the size (diameter) of the individual roots and the extent of crown spread. The dry mass of crown was significantly greater in the 80:20 structural soil treatment while no significant difference was found between the other soil types. The trunk diameter was not significantly different between treatments. Significant differences were observed in the depth of root plates where those grown in top soil had significantly deeper root plates as opposed to the other treatments but though deeper, the vast majority of trees planted in top soil exhibited fibrous rather than structural roots. The cross-sectional area of roots which is indicative of the size of the individual roots showed a significantly greater value in the 80:20 structural treatment while the 50:50 structural and top soil treatments had the lowest values. Significant differences in BM_{max} were only observed in the in situ soil type while the rest of the planting substrates exhibited values that were comparable and not significantly different.

Keywords Structural soil · Tree pulling · Foliar characteristics · Root plate characteristics · Tree stability

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L. F. Ow (✉) · E. K. Sim · Y. K. Fong · P. Y. Tan
Centre for Urban Greenery and Ecology, National Parks Board, Singapore Botanic Gardens, 1 Cluny Road, Singapore 259569, Singapore
e-mail: Genevieve_ow@nparks.gov.sg

F. R. Harnas · I. G. B. Indrawan · A. Sahadewa · H. Rahardjo · E. C. Leong
School of Civil and Environmental Engineering, Nanyang Technological University, 50 Nanyang Avenue, Block N1, Singapore 639798, Singapore

Introduction

Today, Singapore is a jungle of both kinds of concrete and of trees; the tiny island republic of 69,300 ha. is now covered with 1,763 ha. of parks and open spaces, some 3,326 ha of nature reserves, and 4,278 ha of roadside greenery and tree-filled statelands (National Parks Board R.D. 2007). Having effectively covered the island with million of trees over the decades the country has now channeled its efforts into tree preservation and research. As Singapore lies just north of the Equator (Latitude 1.5°N and Longitude 104°E), its geographical location and maritime exposure, lends to a climate that is characterized by uniform temperature and pressure, high humidity and abundant rainfall—some of which have been shown to be relatively disastrous for trees. Furthermore, because of the scarcity of land, trees both large and small are grown in close proximity to footpaths, buildings and roads. Hence, the consequence of tree failure in an urban environment when compared with a rural setting is often more severe. Additionally, since records suggest that storm damage in

the form of stem breakage and uprooting has occurred throughout much of the country (National Parks Board, R.D 2007), and that an urbanized setting does not effectively encourage root growth given space constraints, it is important to explore ways of improving tree anchorage capacity without sacrificing aboveground growth conditions.

Also noteworthy is that the close proximity of skyscrapers in an urban setting often encourages wind-tunneling effects, which can have varying impacts on the stability of trees that should be a cause for concern (Stacey et al. 1994; Gardiner et al. 1997). While there are several options for growing trees in these areas (Smiley et al. 2006), one possible solution to the abovementioned concerns would be the use of a pavement substrate, known as 'structural soil', that has been developed in America and shown to be able to be compacted to meet engineering requirements for paved surfaces, yet possessing qualities that allow roots to grow freely, under and away from the pavement (Bassuk and Grabosky 1996). Structural soils are gap-graded gravels consisting of crushed stone, clay loam, and are generally characterized for their bearing capacity as well as horticultural properties (Fig. 1). The substrate can be compacted to withstand pavement (road surface or sidewalks) loads while permitting root growth. A rigid load-bearing network of stone and soil where the soil fills the macropores formed between the stones provide an integrated, root penetrable, high strength pavement system that extends the growing space of trees to beyond the individual tree pits.

Structural soil is especially important in an urban setting where planting space is limited. Such soil mix ensures adequate amounts of water, nutrients and oxygen will be available for tree roots, providing an ideal soil environment for tree growth and at the same time supporting the pavement load (Bassuk and Grabosky 1996). The additional root space structural soils provide is believed to encourage deep root growth away from the pavement surface. Often, a major impediment to establishing trees in an urban setting is the lack of an adequate volume of soil for tree root growth. Soils under pavements or carriageways are highly compacted to meet load-bearing requirements and

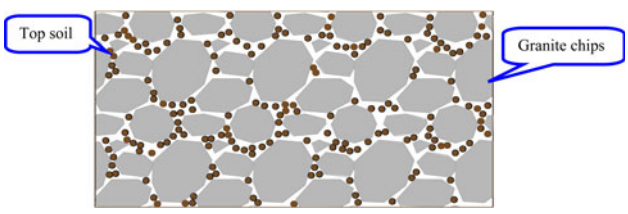


Fig. 1 Structural soils (80:20 mix of granite pieces and top soil, respectively)—a combination of granite chips and top soil to keep the medium root penetrable and support tree growth

engineering standards. This often inhibits root growth causing them to be contained within a very small volume of soil without adequate water, nutrients and oxygen. Previous work estimated that an urban tree lives for an average of 7–10 years, whereas a similar tree in an open/spacious environment can live up to 50 years or more (Grabosky and Bassuk 1995). Additionally, trees that grow within an urban environment often interfere with pavement integrity and have been shown to affect the lifespan of infrastructures substantially.

Given the issues discussed above, structural soil may be a solution for growing trees in the Singapore environment. One significant advantage of structural soil is its ability to meet the load-bearing requirement for structurally sound pavement and carriageway installation while allowing for deep/extensive root growth away from the surface. Highly compacted soils are likely to be impermeable to root growth and water infiltration leaving little available moisture or nutrient holding capacity necessary for growth. The use of structural soil may be able to eliminate this problem to a certain extent.

The work on structural soils have so far been on understanding its impact on the growth characteristics of trees, few studies have studied the benefit of structural soil on the anchorage capacity of trees. Researchers have found a linear relationship between maximum force, stem volume and diameter at breast height (Fraser 1962; Smith et al. 1987; Frederickson et al. 1993; Papesch et al. 1997). Lohmander and Helles (1987), Smith et al. (1987), Valinger et al. (1993), Solantie (1994), Nykanen et al. (1997), Papesch et al. (1997), Ray and Nicoll (1998), Moore (1999), Moore (2000), Peltola et al. (2000), Smiley et al. (2000) and Smiley (2008b) have also shown that the force required to pull a tree to failure is related to the physical characteristics of the tree and the soil type in which it is grown in. The relationship between soil type and tree failure has been repeatedly emphasized in previous reports (Smith et al. 1987; Moore 1999, 2000; Nishimura 2005). This relationship was further supported in an early report by Mergen (1954) who concluded that the distribution and anchoring ability of tree roots are affected by soil texture and consistency. The above statements, however, exclude tree failure via stem fracture which may be partially associated to the root system but more of a result of strong forces (e.g. wind) acting on the tree or on a weak or damage trunk/stem but stem fractures are probably also a result of better/stronger root anchorage which inhibits uprooting which in turn results in the transfer of force up to the tree hence giving rise to stem breakage (Coutts 1983, 1986). The factors determining root cohesion and adhesion to soil were not investigated in this study but such findings will be grounds for future investigation. More importantly, it is noteworthy that in their work with *Picea sitchensis*,

Coutts (1986) observed that the resistance to uprooting was considerably smaller than that of stem breakage, suggesting the need to strengthen weakly anchored trees. This is important as the vast majority of failures in Singapore occur via uprooting on non-cohesive soils making our understanding of tree and soil anchorage a critical aspect in tree care and management. Furthermore, these previous findings provide good evidence to suggest that both soil (texture and structure) and structural properties such as canopy shape, branch structure of a tree and its mechanical stability are closely coupled and the relative strength of the stem and roots will determine the mode of failure.

Information on resistive bending moment from tree pulling forms the basis of mechanistic models such as HWIND, GALES and FOREOLE. Studies based upon these models have shown that uprooting can involve resistance from various sources such as: the strength of windward roots, the weight of the root–soil plate and the strength of the soil at the base of the root–soil plate (Peltola et al. 2000). For example, in their work with Stika spruce (*P. sitchensis*), Ray and Nicoll (1998) concluded that a tree will fail if its uprooting moment exceeds its maximum resistive bending moment. This happens when the center of gravity moves beyond the hinge point in the root system (Ray and Nicoll 1998). The uprooting moment is, however, resisted by the bending of the stem and by various components of root anchorage (as mentioned above) and the mode of failure is dependent upon the strength of the stem and roots.

The wind forces acting on tree stems have been the subject of many studies (Petty and Swain 1985; Milne 1995; Wood 1995; Gilman et al. 2008) and most of these works encompass the mechanics of root anchorage within the soil. This is because the root–soil plate is a compound structure hence the material properties of the soil are critical when determining the overall root anchorage strength (Mattheck et al. 1997). Therefore, the objectives of this paper were (1) to identify the differences in tree stability among trees grown in structural soil and trees grown in natural soil types, working on the hypothesis that structural soil will encourage a more wind firm root system, (2) compare the root architecture of trees grown in the different soil types.

Materials and methods

Experimental trees and site conditions

This experiment was designed to study the forces required to pull trees to failure in natural soils versus trees grown in structural soils. In order to achieve this, a winch and cable system was used to simulate wind loads to trees. The tree-pulling experiment was conducted in March 2009 using 20 rain trees (*S. saman*). The trees were initially planted as seedlings in 2005 in a nursery then transplanted to the experimental plot in 2006. The site was of medium fertility but had a high water table and was often waterlogged so no irrigation was applied. Trees were winched over during dry weather (February–March) when soil moisture was low (between 9 and 20%). Soil moisture was determined using thetaprobe moisture sensors ML2x (Delta-T devices, Cambridge, UK) at 0.5 and 0.8 m depths. The in situ soil at the site is classified as clay with low plasticity (CL) under the unified soil classification system (USCS) with a low permeability rate of 9×10^{-10} m/s.

Soil treatments

The experimental plot had four different soil types: (1) sandy loam soil with some organic matter (topsoil), (2) 80% granite chips and 20% sandy loam soil (80:20), (3) 50% granite chips and 50% sandy loam soil (50:50) and (4) fine-grained sandy clay (in situ treatment—existing soil on the experimental plot excavated and placed back into the tree pit following planting of the tree). The granite chips used here were sub-angular in shape and graded to a size of between 20 and 40 mm. The structural soil was compacted to approximately 90% Proctor density, a level which has been shown to possess load-bearing abilities and can be used as a continuous medium under pavement and car park lots. All soils used in this work except for the existing/natural soil located (treatment 4; in situ soil) on the experimental site consisted of 25–30% sand, 20–40% loam, and 25–40% clay. Organic matter for treatment 1 (topsoil) was between the range 2 and 5%.

The tree pits measured 2.5 (L) by 2.5 (B) by 1 (H) m. Relative density and compaction tests were conducted under laboratory conditions to yield dry density values at various soil moisture contents (Wang et al. 2005; Dryer et al. 2008). The relative compaction that is defined as the ratio of maximum dry density that can be achieved via field compaction to the maximum dry density that can be achieved through laboratory compaction was necessary for the compaction of the structural soil in the tree pit. The maximum dry densities obtained via standard proctor compaction tests conducted in the laboratory for the 80:20 and 50:50 structural soils were 2.17 and 1.92 mg/m^3, respectively (USCS 2010). The prepared tree pits were compacted to the pre-determined dry density levels using a tamping rammer. The natural soil types were not compacted.

Pulling technique

A winch, cable and pulley system (Ingersoll Rand, NJ, USA) was used to pull the trees at a standard height of 1.3 m (Fig. 2) (Stacey et al. 1994; Wood 1995; Gardiner et al.1997; Tosi 2007). To avoid unnecessary complications in the analysis and interpretation of the pull test, the applied force was exerted horizontally (Neild and Wood 1999). The maximum force applied to uproot a tree or fracture its stem was measured and recorded using a digital dynamometer (Dillon EDX; CA, USA) connected to the winch system which records the load set at 2-s intervals. F_{max} was the maximum force measured in the cable. The displacement rate of the experiment is equal to 6 cm/s.

Crown (foliage) and root characteristics

Widths of crown, trunk diameter and root–soil plate parameters were measured for all trees. Two measurements of crown spread for each tree was determined at right angles to one another. The first is the widest crown spread, the greatest distance between any two points along the drip line of the tree. Following that measurement, the axis of measurement is turned 90° for the second measurement. Lastly, the average of both measurements is taken as the crown spread of each tree. The root–plate dimensions were recorded according to Moore (2000). Four basic root plate dimensions were measured with three root plate width measurements—two parallel to the ground and one perpendicular to it. The root plate width was regarded as the distance from the center of the tree stem to the edge of the central mass of roots and soil. The root plate depth was determined correspondingly by pushing a measuring rod through the root–soil plate after tree failure. The fine root tips and the long, rope-like lateral surface roots, which tend to break off during winching were not taken into account since the function of these roots was assumed to be primarily nutritional. The cross-sectional area (CSA) of roots was determined according to Nicoll and Ray (1996).

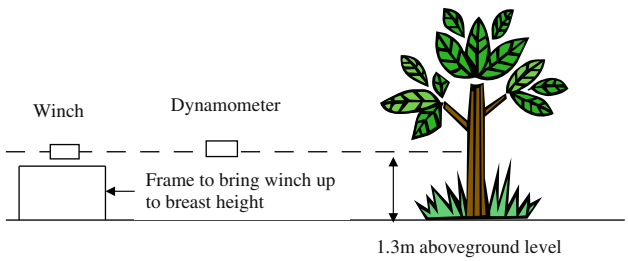


Fig. 2 Schematic diagram showing the setup of the tree-pulling experiment

Data analysis

The maximum resistive moment was calculated from the pull applied and the height of attachment of the pulling rope, which is as follows:

$$BM_{max} = F_{max} \times \text{breast height} \quad (1)$$

where BM_{max} is the maximum bending moment and F_{max} is the maximum applied force (kN) at breast height. The bending moment as a result of the gravitational force induced on the weight of the crown was taken into consideration here through the dry mass measurements of the foliage.

Nested analysis of variance (ANOVA) was used to determine differences in $BM_{max,}$ rooting, stem and crown characteristics among the different soil types (SAS Institute, software Version 9, Cary, NC, USA). Differences were considered significant when $P < 0.05$. Treatment means were compared by least significant difference to determine whether means of the dependent variable were significantly different at $P = 0.05$.

Results

Mode of failure and tree characteristics

Uprooting was the only mode of failure observed in all 20 *S. saman* trees. The uprooting was characterized as the lifting of the intact root plate, with the tree eventually falling under its own weight. Crown mass was not significantly different among trees grown in natural (non-structural soil) soil and those grown in structural soil (Fig. 3a) except for trees grown in the 80:20 structural soil treatments which had a significantly greater mass than all other soil types. The 80:20 structural soil treatment had the greatest trunk diameter but was not significantly different from the other treatments (Fig. 3b).

Root plate characteristics

Root plate depths were significantly different (Fig. 4a). Trees in topsoil had the greatest depth while the root plate in the 50:50 structural soils exhibited the shallowest root plates. Significant differences were also found in the CSA (Fig 4b) of roots grown in the different soil types where roots in the 80:20 structural soil was observed to have significantly greater CSA values. Conversely, the 50:50 structural soil treatments exhibited the lowest CSA value though no significant difference was found between it and the roots grown in top soil (Fig. 4b). Trees growing in the in situ treatment required the most force to pull over (Fig. 4c). There were no significant differences in BM_{max}

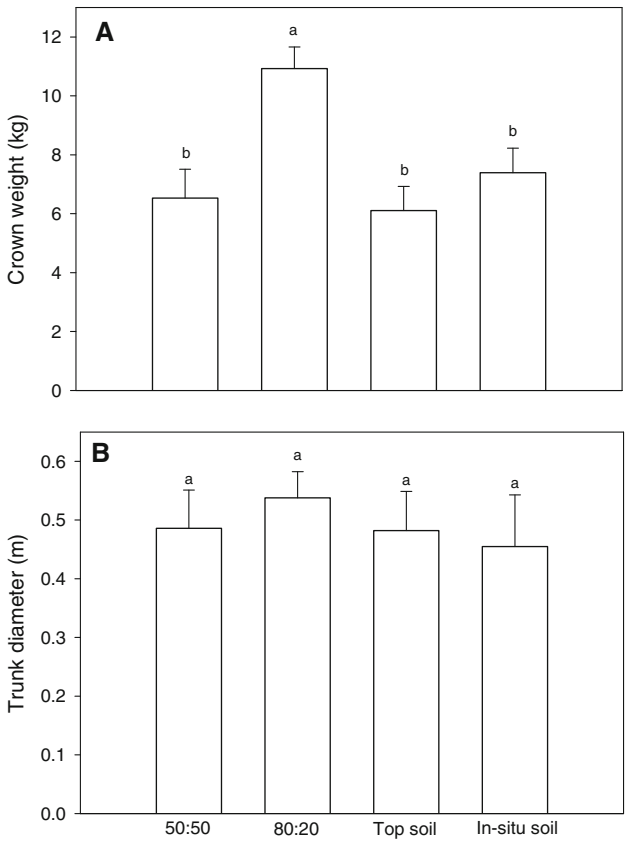


Fig. 3 Comparisons of aboveground characteristics **a** crown dry mass, **b** trunk diameter (DBH) between different soil types. All values are means ± SE; $n = 5$. *Different letters* indicate means are significantly different at $P < 0.05$

observed among the engineered soils and top soil treatments (Fig. 4c).

The fitting of a regression between BM_{max} and crown spread (Fig. 5a) indicated a positive correlation ($R^2 =$ 0.1–0.7). A similar relationship was found between root plate area and BM_{max} (Fig. 5b). An analysis into the CSA of the root system for the different treatments represented in this study showed that a positive correlation exists between CSA and BM_{max} with a R^2 of 0.62 (Fig. 5c). On site observation showed that the root system of trees growing in topsoil and 50:50 structural soil tended to be concentrated towards the direction of the prevailing wind. In contrast, a more even root spread was observed with trees growing in in situ and 80:20 structural soils (data not shown).

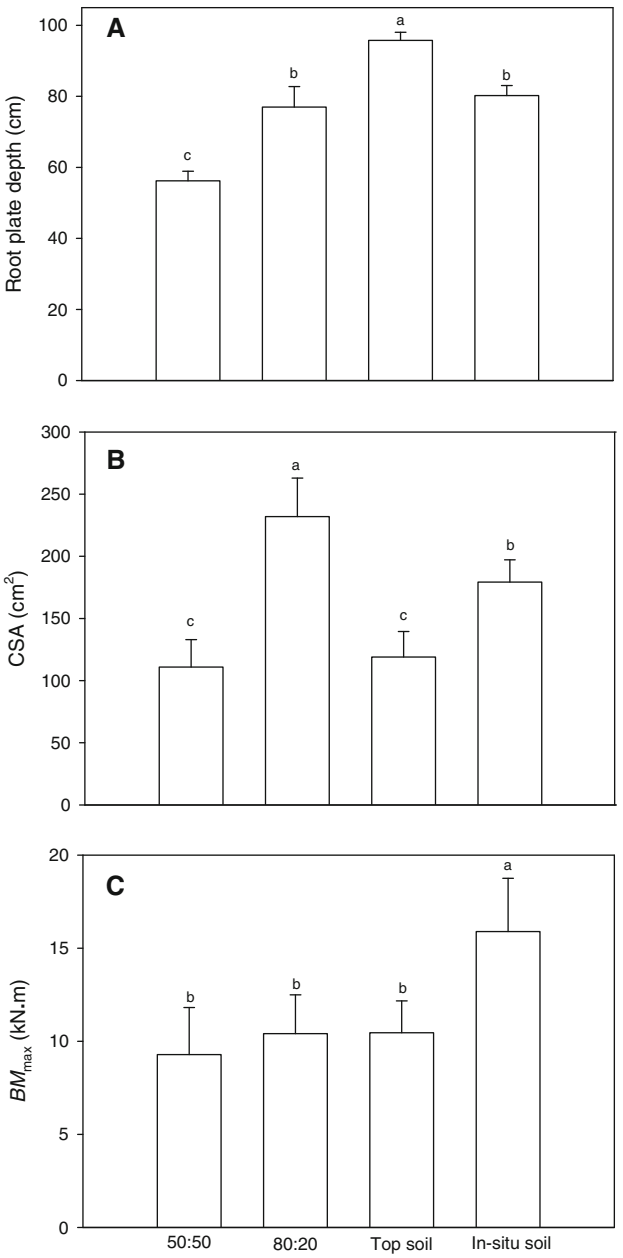


Fig. 4 Comparison of root plate characteristics **a** root plate depth, **b** cross-sectional area of roots (CSA) (cm^2) and **c** maximum applied moment (BM_{max}) between different soil types. Values are means ± SE. ANOVA tests were conducted and *different letters* indicate means are significantly different at $P < 0.05$

Discussion

Structural soil

Structural soil is a planting medium that satisfies the needs and requirements of the horticulturist and engineer. The study of structural soils in Singapore is necessary so that the best alternative to sub-base soil replacement can be established, where the restricted size of a planting pit and the compacted surrounding soils which confines root growth can be altered to provide sufficient soil and rooting volume. Grabosky et al. (1998) established that the use of structural soils allow for the growth of roots beneath paved surfaces and at the same time, encourage root growth away from the paved areas, thus reducing the potential for heaving. Smiley (2008a) has, however, in his study with

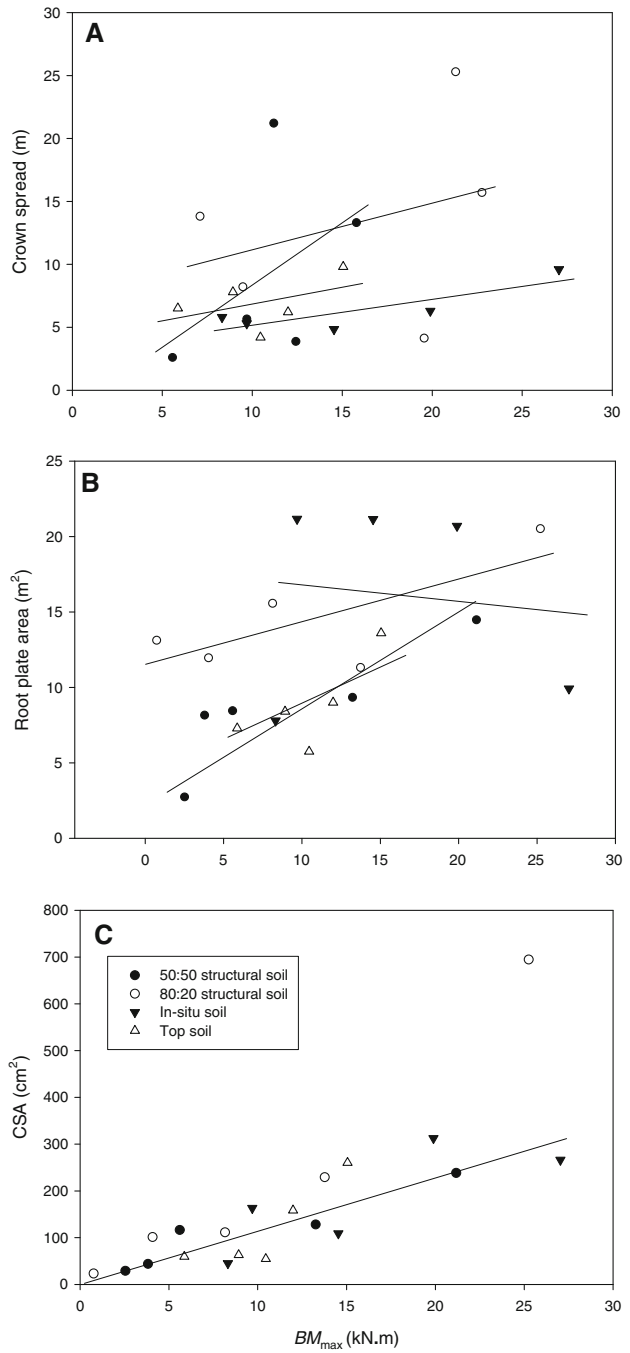


Fig. 5 Relationship between **a** BM_{max} (kN m) and crown spread between the different soil types with the following regression equations. 50:50 structural soil: $y = 0.9799x - 1.4609$ ($R^2 = 0.22$); 80:20 structural soil: $y = 0.3642x + 7.5323$ ($R^2 = 0.11$); in situ soil: $y = 0.2049x + 3.1133$ ($R^2 = 0.70$); topsoil: $y = -0.2712x + 4.0642$ ($R^2 = 0.20$), **b** relationship between BM_{max} (kN m) and root plate area between the different soil types with the following regression equations. 50:50 structural soil: $y = 0.469x + 4.2398$ ($R^2 = 0.78$); 80:20 structural soil: $y = 0.2857x + 11.485$ ($R^2 = 0.54$); in situ soil: $y = -0.1086x + 17.872$ ($R^2 = 0.02$); topsoil: $y = 0.6381x + 2.1352$ ($R^2 = 0.55$) and **c** relationship between BM_{max} (kN m) and CSA (cm^2) with a single linear regression of $y = 14.638x$; $R^2 = 0.62$

London plane trees (*Platanus* × *acerifolia*) found that roots growing in structural soils installed in a 10 cm horizontal layer beneath the pavement, grew well and achieved the largest root diameter of all treatments but also resulted in the largest number of cracks in the pavement. Differences between the two studies are most likely related to the depth of structural soil installation. The use of structural soils in a layer of only 10 cm thick is seldom recommended. Typical installations should be no less than 60 cm in depth (Bassuk and Grabosky 1996).

Static tree pulling

Although tree pulling, is known to overestimate the force required to uproot trees (Tosi 2007), it does, however, allow for the measurement of root plate strength in a consistent manner. Importantly, tree-pulling experiments allow for an understanding of the relationship between root strength and tree stability. The depth of the root plate observed here exhibited some variability between the various treatments, with topsoil exhibiting the greatest depth while 50:50 structural soils had the lowest value (Fig 4a). The data (Fig 4b) presented here for CSA showed that the greatest values were observed in the 80:20 structural soil treatment followed by in situ soil while the 50:50 structural soil treatment had the lowest CSA values though no significant differences was observed between the 50:50 and top soil treatments. The greatest CSA observed in the 80:20 treatments (Fig. 4b) did not, however, correspond to higher BM_{max} values (Fig. 4c). This discrepancy may in part be the result of the extent (spread) of the structural roots or the cohesion and adhesion to the growing substrate or both, which in turn determines the root tensile breaking force and the root tensile strength. Previous work with three shrub species (*Rosa canina*, *Inula viscosa* and *Spartium junceum*) growing on steep hillslopes has shown that the tensile force increases with increasing root spread and size (Tosi 2007). The parameters of root tensile force and strength were, however, beyond the scope of this paper but is well worth further investigation.

Trees winched over top soil though possessing greater depths were characterized by the absence of large structural roots but by the presence of fibrous roots. This has potential implications for the anchorage capacity of the tree but was not translated into significantly lower BM_{max} values as opposed to the structural soil mixes (Fig. 4c). In general, the root plates of trees winched over on in situ and the two structural soils (50:50 and 80:20) were characterized by the presence of structural roots but the BM_{max} values of the two structural soil treatments did not significantly improve despite exhibiting more structural roots. The possibilities relating to this observation and the previous involving

root development in topsoil can be numerous but one potential explanation could be that the presence of granite pieces in the engineered mix may have to a greater or lesser extent, limited the cohesion and adhesion ability between the growing substrate and the tree root system while the exact opposite applies to roots grown in natural soils. The proportion of granite in both treatments is considerably high but necessary as the granite form the load-bearing element of the soil and they can be compacted to avoid settlement. Clearly, the 30% (Treatment 2: 80% granite and treatment 3: 50% granite) difference in granite content had little impact in BM_{max} values, with a 10% difference separating the two structural soil treatments suggesting that rather than the proportion of granite within the substrate, it may be the mere presence of granite chips within the engineered soil that have implications on tree stability. In general, the conclusions that can be derived from Fig. 4a–c are that root architecture and type of planting substrate may be more important than root plate size in defining the overall root anchorage strength (Peltola et al. 2000; Achim et al. 2005).

As with Stokes et al. (1995) and Moore (2000), the observation of larger structural roots developing mostly in the prevailing wind direction here in this study suggests that wind action stimulates the growth of structural roots which are critical for anchorage. Furthermore, there is evidence in the literature to suggest that structural roots may be able to optimize their shape to resist applied loads by concentrating material in the regions of greatest stress (Nicoll and Ray 1996) although this was not confirmed here. Nonetheless, it is evident from recent as well as previous reports (Coutts 1983; Milne 1995; Moore 1999, Moore 2000; Parr and Cameron 2004) that the strength of root anchorage influences the mode of failure. Additionally, Coutts (1986) and Nicoll and Ray (1996) suggested that the weight of the root–soil plate which is a critical component of root anchorage can have a substantial impact on tree stability. This finding was, however, absent from this study as in practice, it is very difficult to do so even if undertaken roughly—in most cases, there was soil dropping off the root–soil plate and many roots tended to break during the pull. It would, therefore be unrealistic to include these data.

Though external loading tends to have a larger effect on tree stability than do self-weight, it will be inappropriate to rule out the role of the tree crown in tree stability and tree failure altogether as crown shape as a result of differential growth responses to the given environments can be in some case the most relevant factor impacting on tree stability. Most trees, however, have a conservative architecture with respect to self-weight, because tree growth tends to comply with mechanical stability criteria (James et al. 2006). The crown weight data in this study showed that a significantly greater crown mass was observed in the 80:20 structural soil treatments which may be attributed to an extensive root system [the presence of structural roots (Fig. 4b)—significantly greater than all other treatments] and this has translated into greater trunk diameter—an architectural strategy conforming to the morphological traits of thicker (and sometimes shorter) trunks and roots with larger diameters to support the heavier crowns (Fig. 3a). This is an important finding as it suggests that despite having a low soil composition and a relatively high granite composition (void of any nutrient and mineral content) the 80:20 mix, did not negatively affect the aboveground growth and development of the trees grown in this particular substrate. The correlation between BM_{max} and crown spread was generally positive in all treatments (Fig. 5a) but this does not rule out the possibility that other parameters apart from crown characteristics may too have a role in determining the stability of a tree. The relationship between BM_{max} and root plate area in the various treatments (Fig. 5b) was generally positive and this was further reinforced by the correlation between CSA and BM_{max} (Fig. 5c) suggesting that the extent of the root plate and the size of individual roots can have implications on the stability of a tree (Nicoll and Ray 1996; Bruchert and Gardiner 2006; Robert and Lindgren 2006). The root plate area is significant in terms of stability since it influences the amount of contact between tree and soil. Generally, a direct link exist between root plate depth and area, the shallower root systems tend to have greater areas and rely more on their rigidity than on their root mass for stability and that this requires stronger structural elements since bending resistance is more critical. The above was observed in all three treatments with the exception of the 50:50 structural mix.

A decline in root strength may reduce the total resistive turning moment of a tree increasing the risk of overturning (Parr and Cameron 2004). A large number of small roots will offer less resistance to bending than a few large roots. Any reduction in the proportion of large roots in relation to overall root biomass would likely have a negative effect on tree stability. This was in contrast to the findings observed here where the root plates developing in the 80:20 structural mix had a significantly greater CSA (indicative of individual roots of larger diameters) value but this was not translated into higher BM_{max} values which may have been the result of poor cohesion/adhesion between the root and the structural soil mix given the high composition of granite chips. Conversely, highly fibrous root plates grown in top soil exhibited similar BM_{max} values to those observed in the 80:20 structural mix.

Conclusion

More importantly, the conclusions that can be drawn from this study are (a) that both crown and root characteristics can have implications on the anchorage and stability of a tree regardless of soil type (e.g. structural soil or natural soils), (b) the establishment of trees within the structural soil substrate (50:50 and 80:20) did not negatively effect the growth performances of the tree, (c) and that possessing an adequate root plate size or large structural roots does not necessarily equate to better anchorage and subsequent stability of a tree.

Importantly, the findings from this study suggest that the use of structural soil as a planting substrate maybe a viable option but in terms of tree stability, the cohesive and adhesive forces between the roots and structural soil mix may not be as effective as oppose to the forces present in natural soil types. In an urban setting where space is limited and trees are grown alongside infrastructures, the need for strongly anchored trees become even more important. As such, future research-related innovation may consider the study of products such as Silva Cells™, which consists of 100% natural soil as an alternative to structural soils.

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References

Achim A, Ruel JC, Gardiner BA, Laflamme G, Meunier S (2005) Modelling the vulnerability of balsam fir forests to wind damage. For Ecol Manag 204:35–50

Bassuk N, Grabosky J (1996) Testing of structural urban tree soil materials for use under pavement to increase street tree rooting volumes. J Arboric 22:255–262

Bruchert F, Gardiner B (2006) The effect of wind exposure on the tree aerial architecture and biomechanics of stika spruce (*Picea sitchensis*, pinaceae). Am J Bot 93:1512–1521

Coutts MP (1983) Root architecture and tree stability. Plant Soil 71:171–188

Coutts MP (1986) Components of tree stability in Stika spruce on peaty gley soil. Forestry 59:173–197

Dryer CL, Kopittke PM, Sheldon AR, Menzies NW (2008) Influence of soil moisture content on soil solution composition. Soil Sci Soc Am J 72:355–361

Fraser AI (1962) The soil and roots as factors in tree stability. Forestry 35:117–127

Frederickson TS, Hedden RL, Williams SA (1993) Testing loblolly pine windfirmness with a simulated wind stress. Can J For Res 23:1760–1765

Gardiner BA, Stacey GR, Belcher RE, Wood CJ (1997) Field and wind tunnel assessments of the implications of respacing and thinning for tree stability. Forestry 70:233–252

Gilman EF, Masters F, Grabosky JC (2008) Pruning affects tree movement in Hurricane force wind. Arboric Urban For 34:20–28

Grabosky J, Bassuk N (1995) A new urban tree soil to safely increase rooting volumes under sidewalks. J Aboric 21:187–201

Grabosky J, Bassuk N, Irwin L, van Es H (1998) Pilot study of structural soil materials in pavement profiles. In: Watson G (ed) The landscape belowground II. International Society of Arboriculture, Champaign

James KR, Haritos N, Ades PK (2006) Mechanical stability of trees under dynamic loads. Am J Bot 93:1522–1530

Lohmander P, Helles F (1987) Windthrow probability as a function of stand characteristics and shelter. Scand J For Res 2:227–238

Mattheck C, Teschner M, Schafer J (1997) Mechanical control of root growth: a computer simulation. J Theor Biol 184:261–269

Mergen F (1954) Mechanical aspects of wind-breakage and windfirmness. J For 52:119–125

Milne R (1995) Modelling mechanical stresses in living Sitka spruce stems. In: Coutts MP, Grace J (eds) Wind and trees. Cambridge University Press, Cambridge, pp 165–181

Moore JR (1999) Effects of soil type on the root anchorage strength of *Pinus radiata* for 'wind and other abiotic risks'. For Ecol Manag 135:20–25

Moore JR (2000) Differences in maximum resistive bending moments of *Pinus radiata* trees grown on a range of soil types. For Ecol Manag 135:63–71

National Parks Board, R.D. (2007) Report on historical analysis of tree failures. National Parks Board, Singapore, pp 1–72

Neild SA, Wood CJ (1999) Estimating stem and root-anchorage flexibility in trees. Tree Physiol 19:141–151

Nicoll BC, Ray D (1996) Adaptive growth of tree root systems in response to wind action and site conditions. Tree Physiol 16:891–898

Nishimura TB (2005) Tree characteristics related to stem breakage of *Picea glehnii* and *Abies sachalinensis*. For Ecol Manag 215:295–306

Nykanen M-L, Peltola H, Quine C, Kellomaki S, Broadgate M (1997) Factors affecting snow damage of trees with particular reference to European conditions. Silva Fenn 31:193–213

Papesch AJG, Moore JR, Hawke AE (1997) Mechanical stability of *Pinus radiata* trees at Eyrewell Forest investigated using static tests. N Z J Forest Sci 27:188–204

Parr A, Cameron AD (2004) Effects of tree selection on strength properties and distribution of structural roots of clonal Stika spruce. For Ecol Manag 195:97–106

Peltola H, Kellomaki S, Hassinen A, Granander M (2000) Mechanical stability of Scots pine, Norway spruce and birch: an analysis of tree-pulling experiments in Finland. For Ecol Manag 135:143–153

Petty JA, Swain C (1985) Factors influencing stem breakage in conifers in high winds. Forestry 58:75–84

Ray D, Nicoll BC (1998) The effect of soil water-table depth on root-plate development and stability of Sitka spruce. Forestry 71:171–182

Robert JA, Lindgren BS (2006) Relationships between root form and growth, stability, and mortality in planted versus naturally regenerated lodgepole pine in north-central British Columbia. Can J For Res 36:2642–2653

Smiley ET (2008a) Comparison of methods to reduce sidewalk damage from tree roots. Arboric Urban For 34:179–183

Smiley TE (2008b) Root pruning and stability of young willow oak. Arboric Urban For 34:123–128

Smiley TE, Key A, Greco C (2000) Root barriers and windthrow potential. J Arboric 26:213–217

Smiley ET, Calfee L, Fraedrich BR, Smiley EJ (2006) Comparison of structural and noncompacted soils for trees surrounded by pavement. Arboric Urban For 32:164–169

Smith VG, Watts M, James DF (1987) Mechanical stability of black spruce in the clay belt region of northern Ontario. Can J For Res 17:1080–1091

Solantie R (1994) Effect of weather and climatological background on snow damage of forests in southern Finland in November 1991. Silva Fenn 28:203–211

Stacey GR, Belcher RE, Wood CJ, Gardiner BA (1994) Wind and wind forces in a model spruce forest. Meteorology 69:311–334

Stokes A, Fitter AH, Coutts MP (1995) Responses of young trees to wind: effects on root growth. In: Coutts MP, Grace J (eds) Wind and trees. Cambridge University Press, Cambridge, pp 264–275

Tosi M (2007) Root tensile strength relationships and their slope stability implications of three shrub species in the Northern Apennines. Geomorphology 87:268–283

USCS (2010) ASTM D2487: 10 standard practice for classification of soils for engineering purposes. In: American Society for Testing and Materials

Valinger E, Lundqvist L, Bondesson L (1993) Assessing the risk of snow and wind damage from tree physical characteristics. Forestry 66:249–260

Wang YH, Zhang JG, Zhang JH, Liu SZ, Wang CH, Xiao QH (2005) Impacts of soil moisture content and vegetation on shear strength of unsaturated soil Wuhan University. J Nat Sci 10:682–688

Wood CJ (1995) Understanding wind forces on trees. In: Coutts MP, Grace J (eds) Wind and trees. Cambridge University Press, Cambridge, pp 133–146